

Problem 1. *Compute the following matrix arithmetic.*

1.

$$\begin{pmatrix} 4 & 7 & 8 \\ 8 & 10 & 0 \\ -5 & 13 & 22 \end{pmatrix} + \begin{pmatrix} 2 & 12 & 3 \\ 6 & 4 & 3 \\ 2 & -9 & 0 \end{pmatrix}$$

2.

$$\begin{pmatrix} 2 & -2 & 2 \\ 6 & 8 & 5 \\ 0 & 23 & 22 \end{pmatrix} + \begin{pmatrix} 2 & 2 & -2 \\ -6 & -4 & -5 \\ 0 & -23 & -18 \end{pmatrix}$$

3.

$$\begin{pmatrix} 5 & 44 \\ 0 & -2 \end{pmatrix} + \begin{pmatrix} 9 & -52 \\ -3 & -4 \end{pmatrix}$$

4.

$$\begin{pmatrix} 22 & 1 \\ 7 & -6 \end{pmatrix} - \begin{pmatrix} -1 & 5 \\ 3 & -2 \end{pmatrix}$$

5.

$$\begin{pmatrix} 21 & -5 & 3 \\ 6 & 10 & 6 \\ 0 & -7 & 12 \end{pmatrix} - \begin{pmatrix} 12 & -14 & -6 \\ -3 & 1 & -3 \\ -9 & -16 & 3 \end{pmatrix}$$

Problem 2. *Matrix Multiplication*

1.

$$\begin{pmatrix} 4 & 7 & 8 \\ 8 & 10 & 0 \\ -5 & 13 & 22 \end{pmatrix} \times \begin{pmatrix} 2 & 1 & 3 \\ 6 & 4 & 3 \\ 2 & -4 & 0 \end{pmatrix}$$

2.

$$\begin{pmatrix} 5 & 0 & 1 \\ 0 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix} \times \begin{pmatrix} 5 & 0 & 1 \\ 0 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix}$$

3.

$$\begin{pmatrix} 5 & 21 \\ 0 & -2 \end{pmatrix} \times \begin{pmatrix} 9 & -5 \\ -3 & -4 \end{pmatrix}$$

4.

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \times \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

5.

$$\begin{pmatrix} 4 & -5 & 3 \\ 6 & 10 & 2 \end{pmatrix} \times \begin{pmatrix} 9 & -1 \\ -3 & 1 \\ -9 & 0 \end{pmatrix}$$

Problem 3. *For parts 1-3 Find the matrix X , for parts 4-5 Find the inverse of A*

1.

$$\frac{3}{2}X + \begin{pmatrix} -1 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 3 & -4 \\ 5 & 4 \end{pmatrix}$$

2.

$$X + \begin{pmatrix} 5 & 2 & -1 \\ -9 & 0 & -5 \\ 0 & 3 & -8 \end{pmatrix} = \begin{pmatrix} 8 & 2 & -4 \\ 12 & 3 & 0 \\ -7 & 3 & -8 \end{pmatrix}$$

3.

$$X \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

4.

$$A = \begin{pmatrix} 5 & 4 \\ 7 & -5 \end{pmatrix}$$

5.

$$A = \begin{pmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{pmatrix}$$

Problem 4. *Some more matrices*

1. *A and B are two matrices. Find their products $A.B$ and $B.A$ - Comment on whether A and B are commutative.*

$$A = \begin{pmatrix} 5 & 3 & 2 \\ 2 & 4 & 1 \\ 8 & 0 & 6 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & -4 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 0 \end{pmatrix}$$

2. *Find the determinants of the above matrices $|A|$ and $|B|$
Show that*

$$|AB| = |A||B|$$

3. *if C is another matrix given by*

$$C = \begin{pmatrix} 2 & 0 & 5 \\ 0 & 2 & 3 \\ 1 & 8 & 4 \end{pmatrix}$$

compute

$$A(B + C)$$

and

$$AB + AC$$

Does the relation

$$A(B + C) = AB + AC$$

remain valid?

4. Compute the transpose of A , B and $(A+B)$. Denoted as

$$(A + B)^T$$

$$A^T$$

$$B^T$$

Does the following relationship hold?

$$(A + B)^T = A^T + B^T$$

5. Consider the transpose of the product of A and B , and demonstrate that the following relationship holds

$$(AB)^T = B^T A^T$$

6. Calculate the product of all 3 matrices ABC

Problem 5. Some more matrices

If A is a matrix, the inverse of that matrix is denoted by A^{-1} and satisfy the condition

$$AA^{-1} = A^{-1}A = I$$

where I is the identity matrix

1. Find the inverse of X where

$$X = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$$

2. Show that

$$AA^{-1} = I$$

and

$$A^{-1}A = I$$

3. Find the eigenvalues and eigenvectors of the following 2×2 matrix

$$\begin{pmatrix} 11 & 10 \\ 1 & 2 \end{pmatrix}$$

4. Find the eigenvalues of the following 3×3 matrix (no need to find the eigen vectors)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 5 \\ 0 & 4 & 3 \end{pmatrix}$$

Problem 6. Some Applications

Consider the 2×2 matrix R

$$R = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Find the following

1. The transpose matrix R^T
2. The inverse matrix R^{-1}

R is known as the rotation matrix, if we want to rotate a vector (x, y) by an angle θ in the anti-clockwise direction, to obtain (x', y')

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

1. If vector $A = (2, 3)$ is rotated by 90° , counter-clockwise, find the coordinates of the resultant vector
2. If vector $A = (2, 3)$ is rotated by 90° , in the clockwise direction, find the coordinates of the resultant vector
3. Show that if A is rotated by 180° , in both clock-wise and counter-clockwise directions, the resultant vector arrives at the same point